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Taxi-Out Time Estimation

Team strong-crane

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Abstract

The taxi-out time of a flight, the time from leaving the stand to taking off, is a basic measure of airport efficiency. The PRC Data Challenge 2026 asks for it at ten European airports, for the 670,790 departures of January, February, June and July 2026. Take-off time, schedule and the NM flight-plan clocks are given; the off-block time recorded by the airport is to be predicted. We treat the task as reconstructing a missing timestamp from several disagreeing clocks in multiple steps: a ridge regression on the NM clocks and ADS-B ground evidence gives a base estimate, and gradient-boosted residual models learn what the clocks miss. Two findings shape everything: First, the 1–2% of departures without an NM record carry a quarter to a third of the squared error, and at Rome their label is one of three recording regimes (stamped at the schedule, an ordinary taxi, or a day-shifted clock). Second, ADS-B ground coverage differs strongly between airports and months. Over the versions of the model the ranking-set RMSE of the pipeline fell from 243.5 to 229.1 s (227.2 s with additional layers); a score-weighted ensemble of our scored submissions, calibrated on the ranking-set scores, reached 212.8 s, and 211.7 s once each day's predictions are reconciled with EUROCONTROL's published daily pre-departure delay. Our final submission uses the pipeline for February and June and that ensemble for January and July (Section 6.5). The code of the pipeline, from the feature table to the submission, is published in the GitHub repository `Somnathab3/strong-crane`; the ADS-B clock files and the feature table are inputs supplied as files.

Keywords: taxi-out time; airport operations; NM flight-plan clocks; ADS-B; gradient boosting; recording regimes; airport hotspots

Abbreviations: ADS-B: Automatic Dependent Surveillance–Broadcast; AOBT: actual off-block time; EOBT / IOBT / LOBT: estimated / initial / last off-block time; NM: EUROCONTROL Network Manager; OPDI: Open Performance Data Initiative; RMSE: root mean squared error

1. Introduction

Taxi-out time drives fuel burn on the ground, runway throughput and the punctuality of a whole rotation, and has been studied extensively with queueing models [13], reinforcement learning [6] and tabular learners on traffic and weather [7, 12, 17], approaches surveyed by Yin et al. [20]. Those studies forecast before pushback from one consistent record, with the stand-to-runway path known. In this challenge the take-off time is known and the off-block time is hidden, so the question is inverted: how well can a missing timestamp be reconstructed from the clocks that are given? The label is a difference between two recording systems, and much of the difficulty is in how each airport records its off-block, not in how aircraft move.

2. Data

2.1 The challenge data

The organising team at the Performance Review Commission of EUROCONTROL provided the movement records of ten European airports [10]: one row per departure or arrival, with 30 fields (airport, runway, stand, aircraft type, operator, flight number, scheduled, take-off and block times, the Network Manager (NM) flight-plan clocks, and the taxi time). The 2025 training set covers 1 January to 31 December 2025 and holds 4,167,797 movements: 2,085,047 departures, each labelled with its taxi-out time, and 2,082,750 arrivals with their taxi-in time; 22,424 departures (1.1%) have no NM record (after duplicated movements are removed; an NM record is an IOBT present). The 2026 ranking set covers January and July 2026 and holds 689,534 movements: 344,841 departures, whose taxi-out time and recorded off-block time are withheld and are to be predicted, and 344,693 arrivals, whose taxi-in time is given; 5,283 of its departures (1.5%) have no NM record. For the final phase the organisers added February and June 2026: the final ranking set holds 1,341,197 movements, 670,790 departures (152,719 in January, 144,158 in February, 181,791 in June and 192,122 in July) and 670,407 arrivals. Its January and July departures are the 344,841 departures of the ranking set,

with the same identifiers; 3,787 of the 325,949 February and June departures (1.16%) have no NM record. The taxi-out label is short for almost every departure (median 15 min, 99th percentile 39 min), but a long tail reaches 36 h (Figure 1), and the departures without an NM record, about 1% of the rows, carry a quarter of the squared error of the pipeline over 2025 (Table 3) and a third on the July 2025 holdout (Table 9).

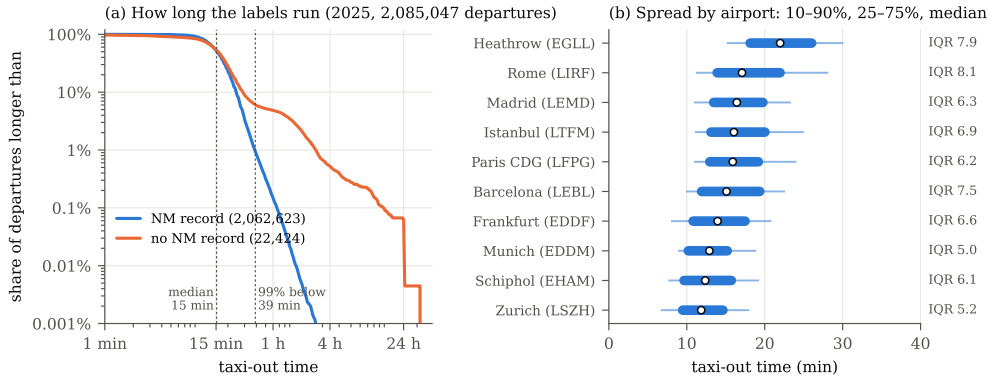


Figure 1: Taxi-out labels in 2025. (a) Share of departures whose taxi-out exceeds a given time (both axes logarithmic), for departures with and without an NM record: the two agree up to the median and part at about 40 min, where departures without an NM record carry a long tail of labels that are hours or a day long. (b) Spread at each airport: 10–90% range (thin line), 25–75% (box) and median (dot), with the interquartile range in minutes; medians run from 12 min at Zurich to 22 min at Heathrow.

2.2 Data types and sources

Besides the challenge data, the pipeline reads six open sources (Table 1). One further source, the free ADS-B Exchange sample of the first day of each month [1], is not under an open licence; it was used only for a correction layer on the 1 January and 1 July 2026 ranking rows of our submissions of early October (7,440 rows, 2.2% of the ranking set), and reaches the final submission only through those files' weights in the score-weighted ensemble. EUROCONTROL's daily punctuality dashboard [8] was used to check the predictions, not as an input.

Table 1: Data types and sources. Coverage is the share of departures with the source's features in the feature tables of the runs (2025 / January + July 2026 / February + June 2026).

No.	Data type	Source	Licence	Coverage (%)
1	Clocks, schedule, traffic	challenge data [10]	challenge	**100 / 100 / 100**
2	Ground surveillance	adsb.lol [2]	ODbL 1.0	**55 / 62 / 62**
3	Airport geometry	OSM [18], X-Plane Gateway [16]	ODbL, Gateway	static
4	Surface events	OPDI v002 [11]	open data	**45 / 45 / 52**
5	Weather observations	METAR (IEM) [14]	public	**100 / 100 / 100**
6	Weather forecast	Open-Meteo [21]	CC-BY 4.0	**100 / 100 / 100**
7	Network statistics	EUROCONTROL [9]	open data	**100 / 100 / 100**

2.3 NM flight-plan clocks

With an NM off-block clock N the anchor $A = T - N$ satisfies $Y - A = N - B$: the error is the disagreement of two recording systems. How large that disagreement is can be measured directly on the 2025 departures, where both clocks are known (Figure 2). The NM actual off-block time AOBT_3 is the closest clock: it lies within 1 min of the recorded off-block for 21% of departures, within 5 min for 75% and within 15 min for 98%. EOBT_1, LOBT and IOBT are planned times: within 5 min for 44–49% and typically 4–5 min earlier than the recorded off-block. Each airport records its off-block in its own way: at Istanbul AOBT_3 is 5 min earlier at the median and within a minute for only 9% of departures; at Rome it is 2 min later, with a tenth of departures more than 12 min apart.

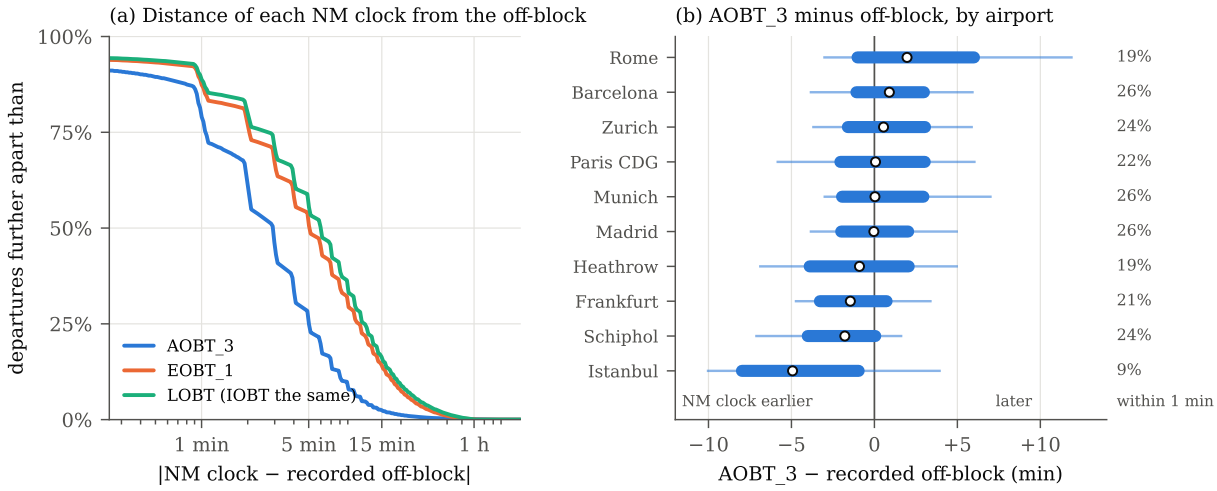


Figure 2: The NM clocks against the recorded off-block time (2025, 2,085,047 departures). (a) Share of departures whose NM clock lies further than a given time from the recorded off-block, for each clock (LOBT and IOBT coincide). (b) AOBT_3 minus the recorded off-block at each airport: 10–90% range (thin line), 25–75% (box) and median (dot), with the share within 1 min at the right.

2.4 ADS-B ground surveillance

Each departure reaches its ADS-B trace through the aircraft's `icao24`, linked from OPDI or from the broadcast callsign (Section 3.2). One aircraft's trace holds its whole day, so the departure's part is cut out of it: points before the aircraft's own in-block belong to the arriving flight and are discarded, and OPDI surface events are taken from the matching flight only (same aircraft, a runway event within 3 min of the take-off, exactly one such flight, and the same callsign), because matching by aircraft alone picks up the inbound flight's events, a median 41 min before the true off-block. ADS-B ground coverage from adsb.lol [2] is the most uneven input (Figure 3). Munich, Amsterdam, Barcelona, and Zurich are heard almost always; Istanbul never; Paris practically never in 2025 and on 8–20% of its departures in 2026; Rome, Heathrow, and Madrid vary by season. In the final ranking set Rome has no ground reception at all in June and July 2026 (0%, against 71–72% in January and February), Frankfurt drops to 26% in February 2026, and Heathrow is heard on 94–98% of its June and July departures, against 18–21% in June and July 2025.

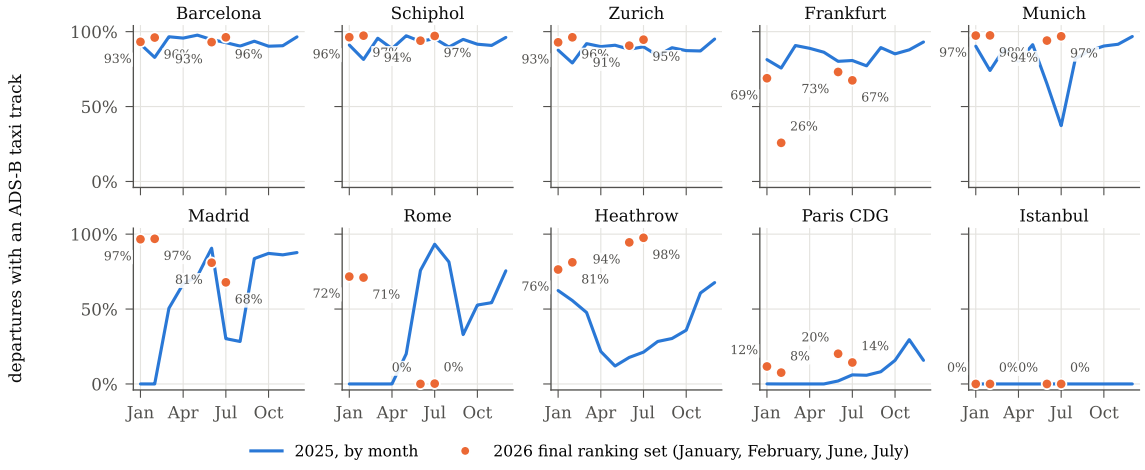


Figure 3: Share of departures with an ADS-B taxi track by airport and month: 2025 (line) and the four months of the final ranking set (markers: January, February, June and July 2026).

Where the aircraft is heard, its track shows the taxi-out directly (Figure 4). SWR758, an A220-300 leaving stand H14 at Zurich on 2 April 2025, left 139 ground reports between 15 min before its off-block and its take-off. The airport recorded its off-block at 15:55:28; the NM AOBT_3 is 32 s later, and the aircraft's first movement of at least 2 kt is 38 s after the recorded off-block. After the pushback it stopped (engine start), taxied west and north to the threshold of runway 10, stopped again before the runway, lined up 681 s after

the recorded off-block and took off at 711 s, which is the label. Fifteen other aircraft moved on the ground in the same eleven minutes. For a departure like this the ADS-B clock and the label agree to within a minute; the error of the model sits in the departures that are not heard, or whose recorded off-block is not a pushback at all.

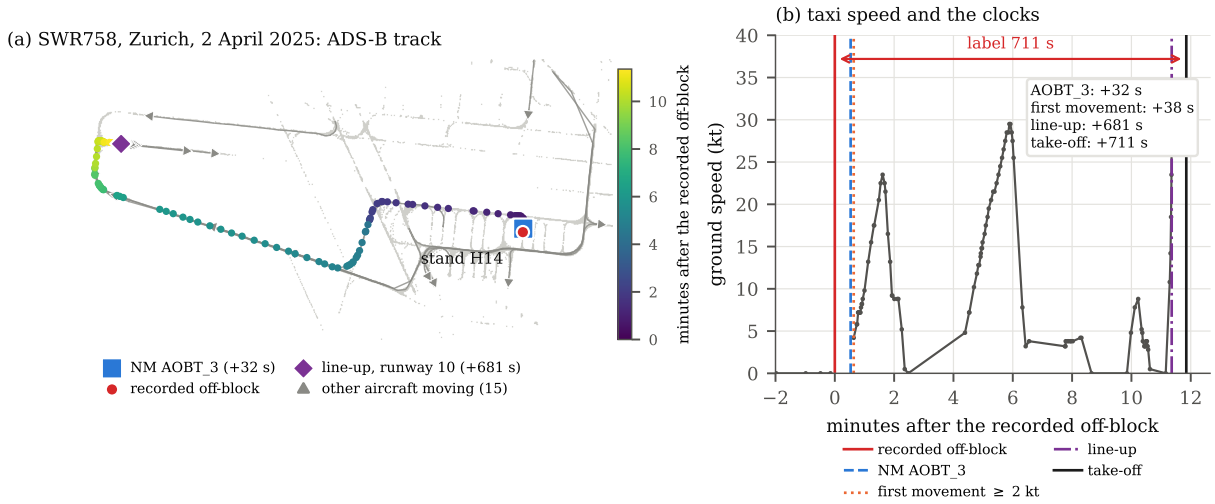


Figure 4: One departure on ADS-B: SWR758, Zurich, 2 April 2025 (A220-300, stand H14, runway 10, label 711 s). (a) Its adsb.lol ground points from the recorded off-block to the line-up on runway 10 (take-off run left out), coloured by minutes after the recorded off-block, over all ground points at Zurich that day (grey, the taxiway layout); the other aircraft moving on the ground in the same period are thin grey tracks, with a triangle at their position and heading when SWR758 lines up. (b) Ground speed from 2 min before the recorded off-block to the line-up, with the recorded off-block, NM AOBT_3, the first movement of at least 2 kt, the line-up and the take-off; the label is the time from the recorded off-block to the take-off.

2.5 Airport hotspots

Where an aircraft stops during its taxi-out is read directly from its ADS-B track: every stop below 2 kt lasting at least 20 s, with its position and duration. A standing aircraft reports rarely, so a gap of up to 15 min between two reports less than 50 m apart also counts as a stop; without this, half of the waiting time is lost. Counting every stop, a departure waits 1.0–3.5 min in total (Figure 5). Most of it is at two ends of the taxi: around the stand after pushback (24–64% of the waiting, the engine start and the clearance to taxi) and at the runway entry (19–52%, the holding point); the taxiways in between hold 11–37%. Rome and Zurich wait mostly at the runway, Schiphol and Munich mostly at the stand, and Madrid more than any other airport on its taxiways.

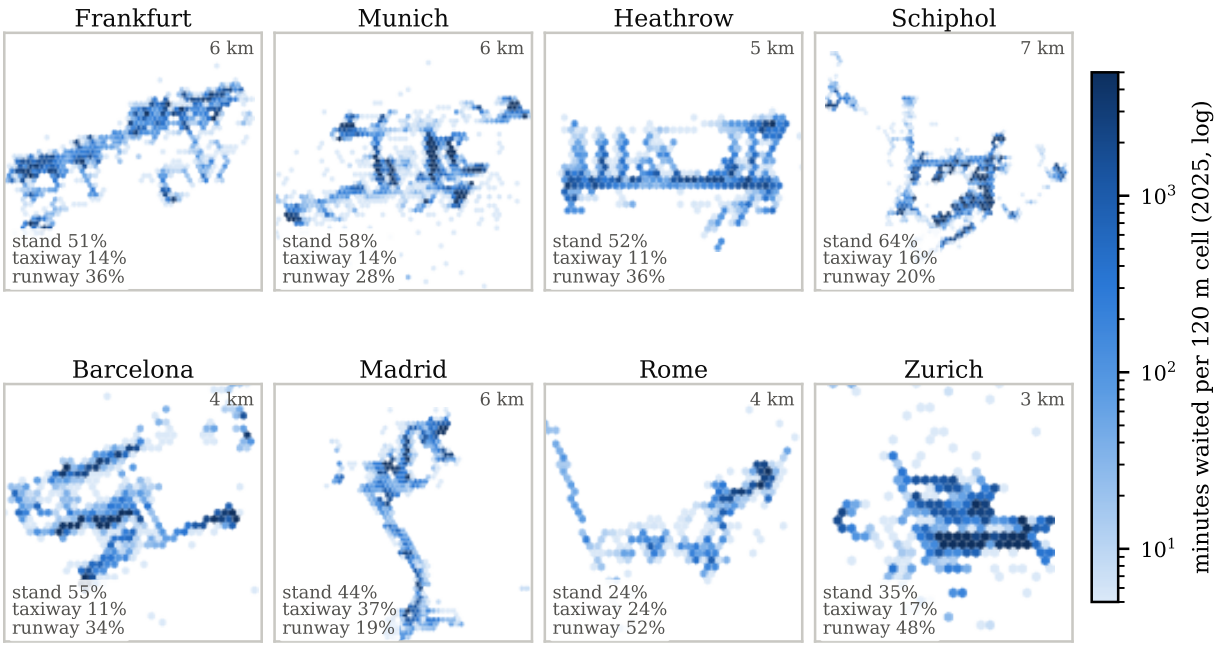


Figure 5: Where departures wait during the taxi-out (2025): minutes spent stopped, summed per 120 m cell (logarithmic scale), from every ADS-B stop of 20 s or more, including stops between sparse reports. Each panel covers every taxi start, stop and runway entry at the airport; the corner gives the share of the waiting at the stand area (within 300 m of where the taxi started), on the taxiways and at the runway entry (within 500 m of the last ground point before take-off). The figure uses its own radii; the model's hotspot features use a 250 m stand area.

The waits at holding points and junctions alone add about a minute of taxi-out time per minute waited (0.8–1.4 min per minute at seven airports; Figure 6). The correlation of the waiting with the taxi-out time stays moderate (0.08–0.47): the waits explain part of the congestion, while most of the variation remains in taxi distance, the recorded off-block and the unseen start of the taxi.

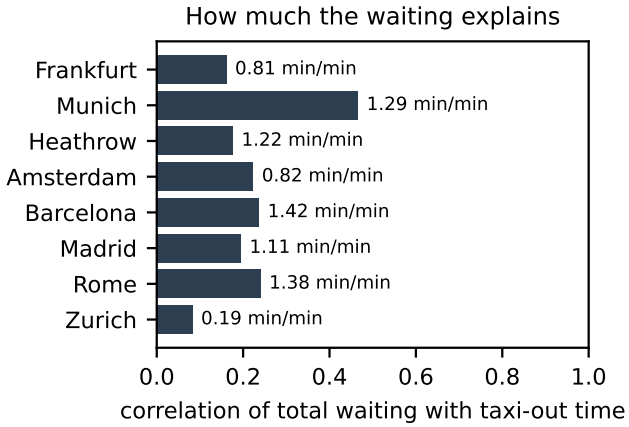


Figure 6: Correlation of the total waiting with the taxi-out time, and the taxi-out minutes added per minute of waiting, by airport (ordinary departures with an ADS-B track, 2025).

2.6 Surface events, weather and network statistics

OPDI [11] matches 45% of the departures to its flight list and, for some of them, gives a taxiway-entry event; it also fills blank aircraft types and operators. METAR (Iowa Environmental Mesonet archive [14]) and Open-Meteo [21] describe the weather at the airport and its change over the previous and next three hours. EUROCONTROL's daily airport tables [9] give the ATFM arrival delay and its causes, pre-departure delay and slot adherence; its monthly taxi-out and taxi-in additional-time tables were tried and dropped, because they drift between 2025 and 2026.

Weather is rare but costly when it occurs (Figure 7). Compared with departures at the same airport in other weather, snow or ice (0.4% of departures) adds 4.6 min to the median taxi-out and 23 min to its 90th percentile, a thunderstorm 3.2 and 11 min, and de-icing risk, a METAR temperature of at most 3 °C with precipitation or fog (1.9%), 2.0 and 13 min; fog and low visibility add about 1 and 5 min, and rain almost nothing. Weather widens the tail more than it moves the median: some flights are de-iced on the way to the runway and others are not. De-icing costs most at Paris and Istanbul (median +4 min, 90th percentile 45 and 57 min) and least at Zurich and Madrid (+0.6 to 0.7 min). The same conditions raise the network delay: on departures with de-icing risk, fog or low visibility the airport's mean ATFM arrival delay that day is about 3.9 min per arrival, against 1.2 min otherwise. Because the conditions are rare, the weather features rank low by correlation over all departures, although they matter for the flights they touch.

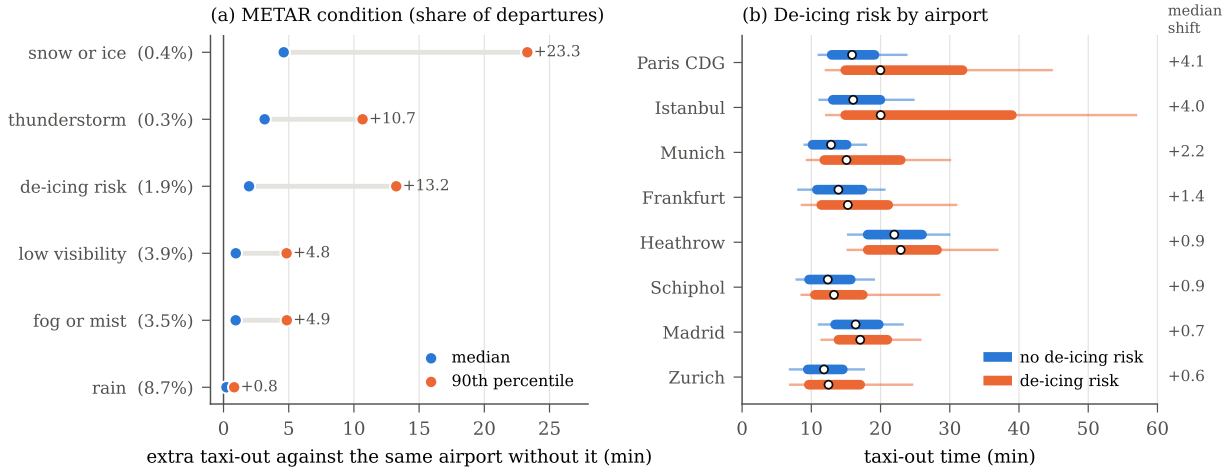


Figure 7: Weather and the taxi-out time (2025 ordinary departures: NM record present, label at most 2 h). (a) Extra minutes at the median and at the 90th percentile under each METAR condition, against departures at the same airport without it (weighted over airports by the number of departures with the condition); the share of departures with the condition in brackets. (b) Taxi-out with and without de-icing risk at the airports with at least 600 such departures: 10–90% range (thin line), 25–75% (box) and median (dot), with the shift of the median at the right.

3. Preprocessing

3.1 Cleaning

Before any feature is built the organiser files are cleaned without reading a label or a block time:

1. the monthly files are merged and duplicated movements (the same movement identifier in two files) are kept once;
2. an NM clock is kept only if the take-off lies between 2 h before and 48 h after it; any other value is treated as missing;
3. runway names are normalised (Rome's "7" beside "07");
4. blank descriptive fields (aircraft type, wake category, market segment, flight type) are filled, in order, from OPDI, the aircraft's own ADS-B record and the same flight number's usual value; each fill is checked on departures with an NM record, where the organiser's own value is known (aircraft type 99% correct from OPDI or the ADS-B record, market segment 92–99%, flight type 99%), and a source flag tells the model whether a value was filled;
5. ADS-B evidence is accepted only inside a physical window: a first point counts as an off-block clock only on the ground (≤40 kt), after the aircraft's own in-block and 30 s to 2 h before take-off, and a pushback signature is dropped if it lies less than 2 min before take-off, is faster than 15 kt, or lies more than 30 min before the aircraft was first heard; ground vehicles are removed from the traces.

3.2 Linking flights to aircraft

The challenge data carry no airframe identifier: no registration (tail number) and no transponder address. Each departure is therefore linked to its aircraft's 24-bit ICAO address (`icao24`) in steps, each tried only if

the one before found nothing:

1. *OPDI flight list* [11]: join on callsign, departure airport and day, which gives the `icao24`, registration and type. If several OPDI flights qualify, keep the one first seen nearest the take-off and carry the number of candidates with it. This links 88–97% of the departures with an NM record (Istanbul 51%).
2. *Broadcast callsign*: in an index of every take-off at the ten airports in the adsb.lol archive, take the nearest take-off within 3 min that broadcasts the departure's callsign.
3. *Airline prefix*: without a callsign, map the flight number's airline to its ICAO callsign prefix (learned on departures with an NM record) and accept the link only if exactly one take-off within 2 min carries that prefix.
4. Otherwise leave the departure unlinked: time alone is never used (4% correct when tested).

The callsign link agrees with OPDI on the aircraft for 96.9% of the departures both link. The `icao24` then connects the departure to its ADS-B trace (Section 2.4); OPDI's aircraft type fills the blank type field above.

3.3 Departures without an NM record

A departure without an NM record has no flight-plan clock and blank descriptive fields. There are 22,424 such departures in 2025 (1.1%) and 5,283 in the ranking set (1.5%), and they occur at every airport (Table 2). By number they are most frequent at Schiphol, Paris, Istanbul and Zurich; their share rose at every airport from 2025 to 2026, most at Schiphol (1.65 to 3.16%) and Zurich (1.70 to 2.85%). The final pipeline does not invent the missing clocks: the blank fields are filled (Section 3.1), the anchor uses the aircraft's ADS-B clock where it was heard and otherwise the airport's median taxi, mixed with the schedule gap by the stamp probability, and dedicated specialists correct the result (Section 5). Rome has fewer of these departures than five other airports, but they are the hardest, because their recorded off-block can be one of four different things (Section 3.4).

Table 2: Departures without an NM record by airport: number and share of the airport's departures in the 2025 training set, the January + July 2026 ranking set and the February + June 2026 months added for the final phase. The February + June column is counted on the final-set feature table, the other two on the ranking-set table.

	2025 training set		2026 Jan + Jul		2026 Feb + Jun	
Airport	Departures	Share	Departures	Share	Departures	Share
Schiphol	4,084	**1.65%**	1,206	**3.16%**	688	**1.83%**
Paris CDG	3,765	**1.57%**	732	**1.84%**	571	**1.52%**
Istanbul	3,623	**1.33%**	719	**1.51%**	608	**1.44%**
Zurich	2,291	**1.70%**	659	**2.85%**	388	**1.78%**
Frankfurt	1,956	**0.85%**	421	**1.16%**	345	**1.00%**
Barcelona	1,856	**1.03%**	441	**1.47%**	317	**1.08%**
Rome	1,487	**0.93%**	383	**1.42%**	236	**0.93%**
Heathrow	1,414	**0.59%**	248	**0.62%**	223	**0.60%**
Munich	1,009	**0.60%**	238	**0.92%**	211	**0.85%**
Madrid	939	**0.44%**	236	**0.64%**	200	**0.56%**
All	**22,424**	**1.08%**	**5,283**	**1.53%**	**3,787**	**1.16%**

The February and June months are closer to 2025 than January and July 2026 are: 1.16% of their departures have no NM record, against 1.08% in 2025 and 1.53% in the ranking set, and the large rises at Schiphol and Zurich in January and July do not recur (Figure 8). Where an NM record exists, all four clocks are present: in every month of 2025 and 2026 the shares of departures with AOBT_3, LOBT, EOBT_1 and IOBT agree with the share with an NM record to within 0.01 percentage points (97.9–99.4%). The question for the final set is therefore not which NM clock is available but whether there is an NM record at all.

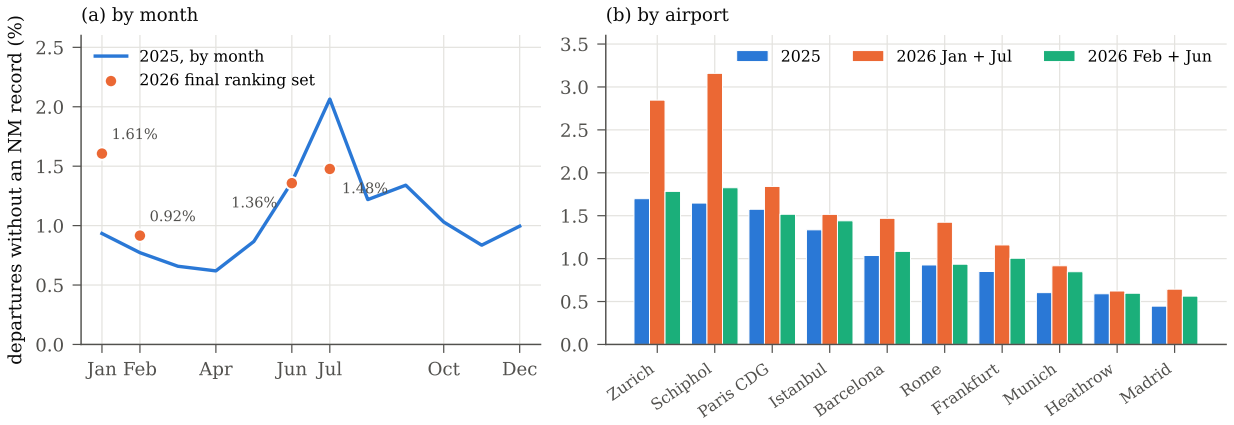


Figure 8: Departures without an NM record in the final ranking set. (a) Share by month: 2025 (line) and January, February, June and July 2026 (markers). (b) Share by airport in 2025, in January + July 2026 and in February + June 2026.

Departures without an NM record are not alike across airports. Among the 16,833 such departures of 2025 that left at least 30 min after schedule, 840 (5.0%) carry an off-block stamped at the schedule (here: label within 5 min of the schedule gap; the stamp classifier and the Rome stage use ±180 s), and 696 of these are at Rome (Table 3). Elsewhere the departures without an NM record are late but ordinary: their taxi-out is a normal taxi and only the flight plan is missing. Their error (218–850 s by airport, against 182 s for departures with an NM record; 2025, each month predicted from the others) comes from the missing clocks, not from stamping; Rome's comes from stamping.

Table 3: Departures without an NM record that left at least 30 min after schedule (2025): number, and how many have an off-block stamped at the schedule; the last column is the share of the squared error of the whole 2025 out-of-fold set carried by all of the airport's departures without an NM record (final pipeline, each month predicted from the others). Counted on the final-set feature table; Figure 9, on the ranking-set table, has one Rome flight fewer.

Airport	Late, no NM	Stamped	Stamped share	Share of all squared error
Rome	1,412	696	**49.3%**	17.1%
Istanbul	1,322	101	**7.6%**	2.6%
Heathrow	1,411	13	0.9%	1.1%
Paris CDG	3,575	16	0.4%	1.1%
Zurich	1,185	5	0.4%	0.7%
Frankfurt	1,822	3	0.2%	0.2%
Munich	973	2	0.2%	0.2%
Schiphol	2,466	2	0.1%	0.5%
Barcelona	1,731	2	0.1%	0.1%
Madrid	936	0	0.0%	0.1%
All	**16,833**	**840**	**5.0%**	**23.6%**

3.4 Rome departures without an NM record

At Rome a departure without any NM clock has a recorded off-block that is one of four things (Figure 9):

1. **The real pushback**, so that the label is an ordinary taxi.
2. **The scheduled off-block time, copied**, so that the label is the whole schedule gap. GTI8237 to Frankfurt on 15 January 2025 was scheduled at 03:15 local time and took off at 07:02; its off-block is recorded at 03:15, the scheduled time, so its label is 13,674 s (3.8 h) against a schedule gap of 13,677 s.
3. **The real pushback clock time written on the schedule's date**, so that the label is 24 h plus a taxi (a day shift). A Jet2 flight to Glasgow (recorded as EXSXS1GT) was scheduled on 24 January 2025 at 11:55 and took off on 25 January at 04:36; its off-block is recorded at 04:24 on 24 January, so its label is 87,177 s, 24 h plus 13 min.
4. **A time-zone shift or another long label**, for example the 1 h UTC offset plus a taxi.

Which one it is depends strongly on the schedule gap: below 2 h most flights have a real pushback (59%),

from 2 to 8 h most are stamped (59–81%), between 8 and 14 h every flight was stamped, and from 14 h on more than half are day shifts.

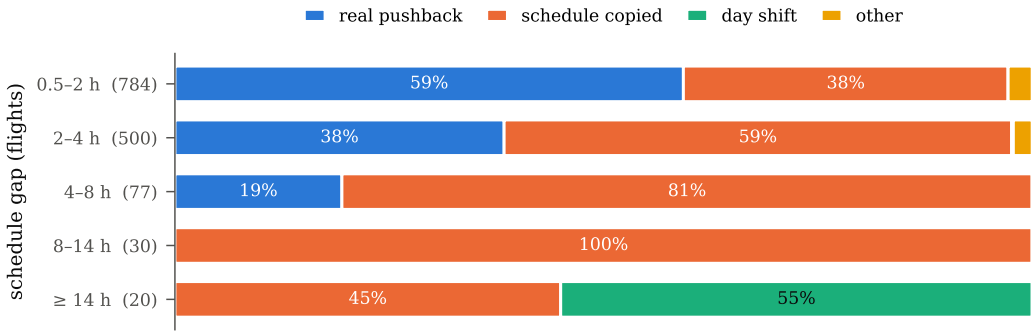


Figure 9: Rome departures without an NM record that left at least 30 min after schedule (2025, 1,411 flights): what the recorded off-block is, by schedule gap (number of flights in brackets).

4. Features

The residual model (model A, Section 5) takes 305 inputs per departure, in eight groups that follow the data types of Section 2 (Table 4). Ten are organiser fields used as they are after cleaning (airport, runway, stand, aircraft type, wake, operator, market, flight type, flight rule, destination); the other 295 are engineered. Five further families, 33 features, were built and dropped because they drifted between 2025 and 2026: context (wind components, runway configuration), EUROCONTROL's monthly taxi-time tables, slowly varying Open-Meteo fields, the inbound aircraft and the seasonal share of departures without an NM record. For the final phase four airport-disruption features, built from the schedule alone, were added (Section 6.5). The hotspot waits are used only at Frankfurt, Schiphol and Barcelona and are missing elsewhere.

Table 4: Model-A features by group: number of features, how many are engineered, and examples.

No.	Group	Features	Engineered	Examples
1	NM clocks and anchor	35	**35**	`nm_a3`, `nm_b`, `csp_a3_minus_eobt`, `anchor`, `e_sched`
2	Flight identity	14	**4**	`airport`, `stand`, `operator`, `stand_runway`, `type_src`
3	Schedule and calendar	18	**18**	`sched_minus_b`, `sched_band`, `hour_sin`
4	Airport traffic	71	**71**	`rb_queue_n`, `arr_done_30`, `stand_turn_age`
5	ADS-B evidence	58	**58**	`cp_rev`, `first_dist_m`, `td_stops`, `hs2_own_hold_s`
6	OPDI events	3	**3**	`op_cp_taxiway`
7	Weather	61	**61**	`mt_is_deicing_risk`, `om_wind_gusts_10m`
8	EUROCONTROL statistics	45	**45**	`ast_atfm_delay_per_arrival`, `ast_slot_late_share`
	Total	**305**	**295**	

4.1 Engineered features by group

1. *NM clocks and anchor* (35): take-off minus each NM clock (AOBT_3, LOBT, EOBT_1, IOBT) and the best available one; the spreads between clocks; which clocks are missing; agreement of the NM flight-plan fields with the movement record; the ridge anchor and its group; and each clock's and the schedule's disagreement with the anchor (`e_*`).
2. *Flight identity* (14): the ten organiser fields, plus the stand–runway pair, the airport–aircraft type pair, and the source flags of the filled type and market (Section 3.1).
3. *Airport traffic* (71): departures, arrivals and runway departures in windows of 15, 30 and 60 min before and after the take-off; the runway queue at take-off (size, age, own rank, headways); completed arrivals and their taxi-in times by airport and runway; the age of the last arrival at the same stand (`stand_turn_age`); wake pairs with the leader; and the taxi-in medians of the stand and runway.
4. *ADS-B evidence* (58): the stand clock, the pushback signature and the first ground point as durations to the take-off, with their distance to the stand, speed and the reporting gap before them; the taxi dynamics (points, span, path, time stopped and moving, stops, turns); and the hotspot waits of Section 2.5.

5. *EUROCONTROL statistics* (45): the day's ATFM arrival delay and its causes, pre-departure delay and slot adherence, with lags, 3- and 7-day means and z-scores, and the destination's ATFM delay.
6. *Schedule and calendar* (18): take-off minus schedule and its logarithm and band; the schedule against the NM clock; malformed or duplicated flight numbers; local hour, weekday and season as cyclic terms.
7. *Weather* (61): METAR observations, condition flags and their changes over 1 and 3 h; Open-Meteo hourly fields; precipitation, snow and gusts over the previous and next 3 h.
8. *OPDI events* (3): whether the flight was matched, the time from its taxiway entry to the take-off, and the number of taxiway entries.

5. Model

The model reconstructs the missing off-block time as a sequence of stages, each correcting what the one before leaves (Figure 10). It starts from a transparent estimate built on the clocks, lets gradient-boosted models learn what the clocks miss, and then treats the departures whose label is not an ordinary taxi separately. Every stage predicts the taxi-out time in seconds; the final prediction is the last stage's output.

5.1 Validation

July 2025, a calendar month of the ranking set, is held out: it is predicted by a stack fitted and tuned on the other eleven months. Inside those months the stack is cross-validated over four contiguous month blocks, never over days, because neighbouring days share flights and airlines. Every choice the stack makes (ensemble weights, which airport gets an expert, a specialist or a gate) is made on out-of-fold predictions and then frozen, and every stacked stage learns from the out-of-fold predictions of the stages below it, so no stage sees its own training labels through another; inside the cross-validation the choices use pooled out-of-fold predictions and read slightly optimistic, while the July holdout does not. Hyperparameters were tuned on the other months only (Section 5.3).

5.2 Stages

The pipeline works in four phases (Figure 10). It first builds a simple estimate from the clocks; then gradient-boosted models learn what that estimate misses; then the departures whose label is not an ordinary taxi get their own models and rules; and finally every prediction is kept inside the range the clocks allow. Table 5 gives each stage in one line.

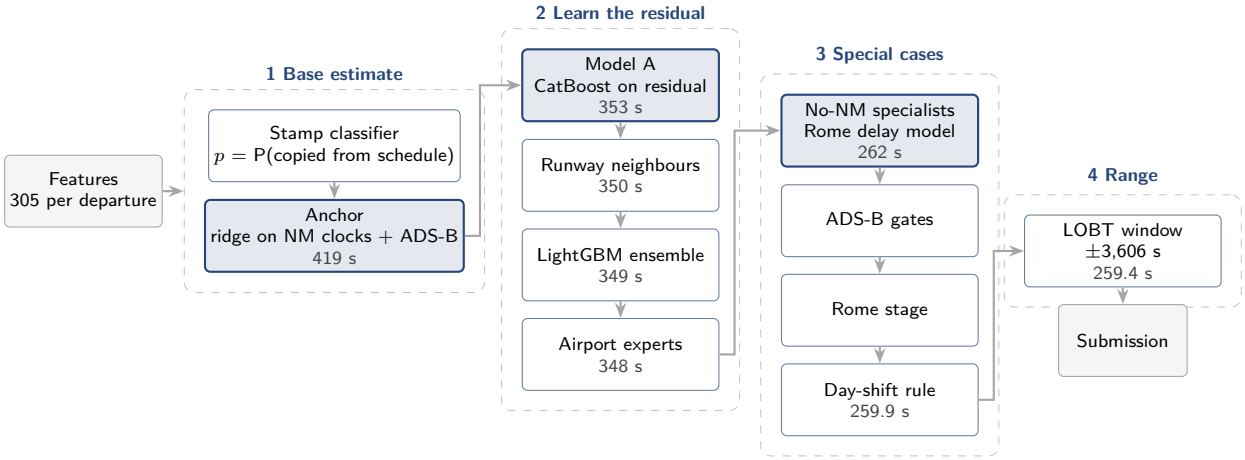


Figure 10: The pipeline. Each box is a stage; the grey number is the July 2025 holdout RMSE after it (Table 7). Shaded boxes do most of the work.

Table 5: The stages at a glance: which departures each one touches and what it does.

No.	Stage	Applies to	What it does
1	**Base estimate**		
1	Stamp classifier	no NM, or schedule far off	P(off-block copied from the schedule)
2	Anchor	all	ridge per airport, runway and ADS-B band on the NM clocks + ADS-B
2	**Learn the residual**		
3	Model A	all	CatBoost on label − anchor, 305 features
4	Runway neighbours	all	how late nearby departures on the runway run
5	LightGBM ensemble	all	a second learner, combined with model A per airport
6	Airport experts	airports with ≥5,000 rows	one model per airport, mixed in where it helps
3	**Special cases**		
7	No-NM specialists	no NM record	own models; at Rome a model of the off-block delay
8	ADS-B gates	ADS-B heard on the ground	correction from the aircraft's own track
9	Rome stage	Rome, no NM, ≥1 h late	stamped or ordinary below 8 h, the gap from 8 to 14 h
10	Day-shift rule	NM, logged >20 h after IOBT	movement time minus IOBT
4	**Range**		
11	LOBT window	all with a LOBT	keep the prediction within 3,606 s of LOBT [3]

The anchor in detail. With an NM clock the anchor is $b + c_0 + c_1 z + c_2 e + c_3 d$. Where all four NM clocks exist, b is a clock ridge: a ridge per airport and runway on the four clock gaps (take-off minus AOBT_3, LOBT, EOBT_1 and IOBT) and three of their spreads; otherwise b is take-off minus the best NM clock (AOBT_3, else LOBT, EOBT_1, IOBT). z is the EOBT_1 gap minus b, e the ADS-B duration minus b and d the distance of the first ADS-B point from the stand. The coefficients are fitted per airport, runway and ADS-B evidence band (pushback seen; first point at the stand; within 1 km; 1–1.5 km; none) where there are at least 200 rows, else per airport and band, else pooled, so the fit itself decides how far each kind of ADS-B evidence is trusted. Without an NM clock the anchor is the ADS-B duration or the airport's median taxi, mixed with the schedule gap by the stamp probability p.

Choices made out of fold. Whether an airport gets an expert, a specialist or a gate, and every ensemble weight, is chosen on out-of-fold predictions only: a specialist must lower the error in at least three of the four folds, with its three largest rows carrying at most 90% of its gain; an expert gets the share (0, 0.25, 0.5, 0.75 or 1) that lowers the out-of-fold error on ordinary labels; a gate stays on unless it hurts (ADS-B coverage changed between 2025 and 2026), and the gates are never applied at Paris, whose first ADS-B points are inbound landings.

5.3 Hyperparameters

Model A, a CatBoost model [19], and the LightGBM learner [15] were tuned with Optuna [4] on months outside the holdout (fitted on February–June and August–October, validated on November–December); the stamp classifier and the gates were searched the same way without gain, and the specialists, experts and the Rome classifier use fixed settings. Table 6 lists what the final pipeline uses.

Table 6: Settings of every learned stage in the final pipeline. CatBoost and LightGBM minimise squared error; the classifiers, log loss.

Stage	Learner	Settings
Stamp classifier	CatBoost	300 trees, depth 6, learning rate 0.1; 3 seeds
Model A, + neighbours	CatBoost	4,100 trees, depth 8, learning rate 0.096, L2 1.0, border count 128, bagging temperature 0.092; residual clipped to ±2 h; 4 seeds
LightGBM	LightGBM	1,650 rounds, learning rate 0.0251, 219 leaves, 41 rows per leaf, feature fraction 0.553, bagging fraction 0.688, L2 10.49
Airport experts	CatBoost	2,000 trees, depth 8, learning rate 0.096, L2 1.0
No-NM specialists	CatBoost	1,500 trees, depth 6, learning rate 0.05, L2 3 (depth 4, L2 10 below 5,000 rows); 5 seeds
NM gate	CatBoost	1,500 trees, depth 7, learning rate 0.08, L2 3; correction clipped to ±30 min
No-NM gate	CatBoost	800 trees, depth 6, learning rate 0.05, L2 3
Rome classifier	CatBoost	400 trees, depth 4, learning rate 0.05, L2 10; 3 seeds

5.4 What each stage achieves

Table 7 follows the July 2025 holdout through the stages, separately for departures with and without an NM record. The anchor alone scores 419 s. Model A removes 66 s, almost all of it on departures with an NM record (293 to 218 s; outside Rome 268 to 193 s). The runway neighbours, the LightGBM ensemble and the airport experts add 1–4 s each. The specialists leave the NM departures unchanged and act only on the 2.1% without an NM record, from 1,931 to 1,090 s, which takes the overall error from 348 to 262 s; at Rome those departures fall from 6,388 to 3,334 s, elsewhere they are already at 510 s and barely move. The gates, the Rome stage and the day-shift rule give the final 259.9 s; the LOBT window, applied afterwards, gives 259.4 s.

Table 7: Holdout RMSE (s) after each stage by population, July 2025 (number of departures in brackets). "Final" includes the ADS-B gates, the Rome stage and the day-shift rule.

		NM record		No NM record		
Stage	All (190,713)	all (186,775)	outside Rome (171,901)	all (3,938)	outside Rome (3,601)	Rome (337)
Anchor	419.4	293.1	268.0	2,108	633.5	6,903
Model A	**353.3**	**217.8**	**193.1**	**1,949**	**529.8**	**6,432**
+ runway neighbours	**349.5**	**214.0**	**188.0**	**1,935**	**509.8**	**6,401**
Ensemble with LightGBM	**349.3**	**213.4**	**187.3**	**1,936**	**511.4**	**6,403**
+ airport experts	**348.0**	**212.3**	**186.2**	**1,931**	**509.6**	**6,388**
+ no-NM specialists	**262.0**	**212.3**	**186.2**	**1,090**	**509.6**	**3,334**
Final	**259.9**	**211.1**	**185.6**	**1,077**	**508.7**	**3,283**

By airport (Table 8) every airport gains most from model A (39–112 s against the anchor), and Istanbul gains another 19 s from its runway neighbours. With an NM record the final error runs from 90 s at Zurich and 106 s at Schiphol to 226–248 s at Heathrow, Barcelona, Paris and Istanbul, where ADS-B is missing or the off-block clock is rounded; Rome's NM departures are at 402 s. Without an NM record every airport is worse, from 196 s at Madrid to 1,144 s at Heathrow and 3,283 s at Rome, so the share of such departures, 1–3% everywhere, decides much of each airport's total.

Table 8: Holdout RMSE (s) by airport after the main stages, July 2025, ordered by the final error.

Airport	Departures	No NM (%)	Anchor	Model A	+ experts	Final, NM	Final, no NM	**Final**
Schiphol	22,489	2.3%	159.1	113.3	111.8	106.2	221.2	**110.2**
Zurich	12,611	2.7%	223.1	131.1	122.7	89.6	497.2	**120.7**
Frankfurt	21,334	2.0%	203.8	137.3	128.9	118.7	308.2	**125.5**
Munich	15,825	1.1%	208.4	154.1	148.6	134.6	585.6	**147.2**
Madrid	18,818	1.5%	246.0	169.5	164.3	162.9	196.4	**163.4**
Barcelona	16,752	2.6%	266.0	227.1	224.8	225.0	236.1	**225.3**
Paris CDG	21,712	3.1%	326.2	239.8	237.7	229.9	414.9	**237.7**
Istanbul	25,171	1.7%	360.0	275.5	256.0	247.8	552.0	**256.0**
Heathrow	20,790	1.6%	387.5	275.3	267.8	225.9	1,144.2	**267.4**
Rome	15,211	2.2%	1,139.8	1,038.0	1,032.1	401.7	3,282.8	**629.7**
All	190,713	2.1%	419.4	353.3	348.0	211.1	1,076.5	**259.9**

A single departure can move these numbers. The largest error of the July holdout is one Rome departure without an NM record, scheduled 16.3 h before its take-off, whose label is a day shift (87,186 s, 24 h plus a taxi); the stack predicted 65,407 s, and this one row carries 3.7% of the month's squared error (called correctly the month would score 255.1 s). The ten largest errors, all at Rome and all without an NM record, carry 13.7% of the squared error, and leaving them out would give 241.5 s.

6. Results and Discussion

The pipeline scores 259.4 s on the July 2025 holdout (Section 5.4). On the 2026 ranking set it was submitted inside our best submission before the final phase (227.17 s), together with layers that are not part of the pipeline described here; refitted on all of 2025 for the final phase and submitted on its own it scores 231.71 s (Table 11). This section asks where the remaining error sits, what the pipeline does against each part of it, and what carried over from 2025 to 2026.

6.1 Where the remaining error is

The error is concentrated in a few populations (Table 9). Departures with an NM record outside Rome are 90% of the rows and are predicted to 186 s; they still carry 46% of the squared error simply by their number. Within them, the ADS-B evidence decides: where the pushback was seen the error is 124 s, where the aircraft was heard only later or not at all it is 222–228 s, because the prediction then rests on the NM clocks and their disagreement with the airport's record (Figure 2). The rest of the error is in departures whose label is not an ordinary taxi. Labels longer than 2 h are 120 departures (0.06%) and 20% of the squared error; Rome's departures without an NM record are 337 (0.18%) and 28%, of which the 157 that left at least 2 h after schedule alone carry 25%. A single day-shifted Rome departure carries 3.7%.

Table 9: Where the final error is: July 2025 holdout (190,713 departures); shares of the squared error of 20% or more in bold. For the ADS-B rows a departure is heard on the ground if any ground point was received (`adsb_any`); the anchor's evidence bands split the same departures differently.

Population	Departures	Share	RMSE (s)	Share of squared error
By NM record and airport				
NM record, outside Rome	171,901	90.1%	**185.6**	**45.9%**
NM record, Rome	14,874	7.8%	**401.7**	18.6%
No NM record, outside Rome	3,601	1.9%	**508.7**	7.2%
No NM record, Rome	337	0.2%	**3,282.8**	**28.2%**
NM departures by ADS-B evidence				
Pushback seen	30,099	15.8%	**123.8**	3.6%
Heard on the ground, no pushback	101,456	53.2%	**221.6**	**38.7%**
No ADS-B	55,220	29.0%	**228.2**	**22.3%**
By label				
Label at most 2 h	190,593	99.94%	**232.4**	**79.9%**
Label over 2 h	120	0.06%	**4,647.2**	**20.1%**

6.2 What reduced the error

Each part of the pipeline answers one source of error (Table 10). The anchor and model A deal with the disagreement between the NM clocks and the airport's record and take the NM departures from 293 to 218 s; the traffic around the departure, through the runway neighbours, the LightGBM ensemble and the airport experts, adds another 6 s. The departures without an NM record need their own models: the stamp probability enters the anchor, and the specialists take these departures from 1,931 to 1,090 s, the largest single gain of the pipeline. Uneven ADS-B coverage is handled by trusting each kind of evidence only as far as the fit allows and by switching the gates off where they hurt. The rules at the end, the day-shift rule and the LOBT window, follow from properties of the timestamps and gained on both the holdout and the ranking set.

Table 10: Sources of error and what the pipeline does against each, with the measured effect (July 2025 holdout unless stated).

Source of error	What we implemented	Effect
NM clocks disagree with the recorded off-block	anchor per airport and ADS-B band; model A on its residual	NM departures 293 → **218 s**
Congestion at the moment of departure	runway neighbours, LightGBM ensemble, airport experts	NM departures 218 → **212 s**
No NM record	stamp probability in the anchor; no-NM specialists	no-NM departures 1,931 → **1,090 s**
Uneven ADS-B coverage, inbound first points	physical window; evidence bands; gates on unless they hurt, never at Paris	pushback seen: **124 s** against 222–228 s without
Rome's recording regimes	Rome stage: classifier below 8 h, schedule gap from 8 to 14 h	Rome no-NM squared error **−10%** (earlier development run, January + July 2025)
Movement logged a day late	day-shift rule (movement − IOBT)	one NM departure in 2025
Predictions outside the clocks' range	LOBT window, ±3,606 s	259.91 → **259.38 s** (development run); ranking 227.51 → **227.17 s**
Drift between 2025 and 2026	33 drifting features dropped; four seeds	ranking 227.96 → **227.51 s** (seeds)
Disruption at the airport (final phase)	share of departures ≥1 h behind schedule in the airport-hour and -day, gap rank; eight seeds	260.69 → **258.87 s** (pipeline rerun on the final-set feature table, four seeds, as baseline); ranking 231.71 → **230.67 s** (in the 0.75 ensemble)
One model family only (final phase)	equal-weight ensemble with the earlier stack, weight fixed in advance	ranking 230.67 → **229.10 s**

6.3 Rome

Rome holds 47% of the July squared error, most of it in departures without an NM record. For these the label is one of the four things of Section 3.4, and a stamped and an ordinary flight with the same three-hour delay have labels three hours apart. Between 8 and 14 h of delay every 2025 flight was stamped, and from 20 h on most were day shifts, so those bands are rules; the error sits in the stamped-or-ordinary decision below 8 h.

In the development repository every signal we could construct was measured against the 2025 labels, each month predicted from the others; the strongest (the flight number's frequency, the carrier's stamp history, the inbound aircraft) rank the flights well, but in the middle of the probability range the two kinds have identical profiles. With a probability p of a stamp the prediction that minimises the expected squared error is the hedge $p \cdot \text{gap} + (1 - p) \cdot \text{ordinary}$; a rule that commits is wrong by the whole gap when it is wrong, so every rule list lost to the hedge out of month, and an oracle knowing each flight's kind would reach about 900 s on these flights against 2,800–3,050 s for a calibrated hedge. Two options measured afterwards and not submitted, a classifier weighted by the cost of a wrong call and the expected value over the day-shift outcomes, take the July holdout from 259.9 to 253.7 s.

6.4 What carried over to 2026

The ranking-set score fell from 243.53 s with the first staged stack to 227.17 s with our best submission before the final phase and, for a submission built only from the pipeline's models, to 229.10 s (Table 11). The submissions before the final phase were ensembles with the earlier stack and carried layers outside the pipeline described here: Rome rules, Rome rows copied from an earlier submission that was partly calibrated on the ranking set, and the ADS-B Exchange first-of-month layer. The steps that helped on the 2025 holdout mostly helped on the ranking set too; the LOBT window, for example, gained on both (July holdout 259.91 to 259.38 s, ranking set 227.51 to 227.17 s). ADS-B coverage also changed: at Heathrow pushback clocks rose from 3% of July departures in 2025 to 72% in 2026, while Paris and Istanbul have none in either year.

Table 11: Ranking-set score (January + July 2026) of the main submissions (s), in order. The earlier stack is the one of the second row. Rows 3 to 6 are ensembles with the earlier stack that carry layers outside the pipeline described here (Section 6.4); the last two rows are the score-weighted ensemble (Section 6.5).

Step	What changed	Ranking set
1	first staged stack	243.53
2	field filling, airport experts, Rome stage (the earlier stack)	**233.04**
3	Rome rules, ensemble with the earlier stack	**228.52**
4	drifting features dropped (one seed), ensemble with the earlier stack	**227.96**
5	the pipeline described here with four seeds, in the same ensemble	**227.51**
6	LOBT window	**227.17**
Final phase (refitted on all of 2025)		
7	the pipeline alone, LOBT window	231.71
8	ensemble: the pipeline with disruption features and eight seeds (0.75) and step 7 (0.25)	**230.67**
9	ensemble: the earlier stack (0.5) and step 8 (0.5)	**229.10**
	score-weighted ensemble of 64 scored submission files	*213.52*
	the same with the later submissions added, step 8 among them	*212.85*
	the same after the post-processing stage (Section 6.5)	*211.96*
	the ensemble recomputed with the post-processed files (the final file)	*211.73*

6.5 The final phase: February and June 2026

For the final set, the 670,790 departures of January, February, June and July 2026, we followed the process described above and ran it end to end on the final organiser files: the pipeline refitted on all of 2025, with four airport-disruption features and eight seeds; an ensemble with the earlier final-set run (weight 0.75 on the new run) and an equal-weight ensemble with the earlier stack; then the LOBT window. For February and June the earlier stack's predictions come from its code rerun unchanged on the final organiser files (July 2025 holdout 263.67 s, against 259.04 s in its first run, because its inputs were rebuilt since). Every added feature and every weight was chosen on the 2025 labels, and on January and July 2026 this model scores 229.10 s (Table 11); the February and June departures of the submitted final file carry it.

For January and July the submitted rows are a score-weighted ensemble of our own scored submission files. Because the official RMSE of each file is known, the error cross-moment of any two files follows from their scores and their difference, so the weights that minimise the ensemble's error are known in closed form; a small ridge keeps them finite where files are near-copies, and the LOBT window is applied afterwards. Its members include files that changed selected rows to read their values from the score. The ensemble of 70 files scores 212.85 s. This is calibration on the ranking set, not a model: its gain sits almost entirely in Rome's departures without an NM record (copying only those 356 rows from an earlier score-weighted ensemble into our best submission before the final phase moves 227.17 to 218.66 s), and it does not carry over to February and June, which hold the pipeline's own predictions. Members, scores, ridge and weights are in the repository's submission manifest, from which the file is rebuilt.

After the ranking phase one post-processing stage was added to the pipeline and applied to both parts of the file. EUROCONTROL publishes, per airport and day, the total pre-departure delay of the day's departures [9]; on the 2025 labels that total equals, to within rounding, $\sum_i \max(0, d_i - y_i)$ over the day's departures, with d_i the gap from schedule to take-off and y_i the taxi-out (correlation 0.9999 with the day of the actual off-block). Three steps use it. Each airport-day whose departure count matches the published count (within five) is reconciled: the predictions of its delayed departures ($d_i > p_i$) are moved by one common amount until the sum matches the total. Every departure of the day then responds to that day shift with a loading, fitted on the 2025 out-of-fold predictions per schedule-gap band and delayed state (0.2 to 0.8 for ordinary departures, more for delayed ones). Finally, a Rome or Istanbul departure without an NM record whose flight has an NM record at its destination airport did not match its plan, so its off-block lies outside the LOBT window of that plan (none of 190 such 2025 labels lies inside it); a prediction inside the window is moved to its nearer edge. On the 2025 out-of-fold predictions the reconciliation takes the error from 230.9 to 229.0 s and the three steps together to 228.6 s, improving ten of eleven months and seven of ten airports; applied to the ensemble's January and July rows the reconciliation takes 212.85 to 212.05 s and the three steps to 211.96 s; the ensemble recomputed with the post-processed files among its members scores 211.73 s. The submitted final file carries the three steps on February and June and that recomputed ensemble on January and July.

6.6 Feature importance

Which features model A relies on is measured with CatBoost's prediction-value change, averaged over the four seeds of the final pipeline and normalised to 100%. The importance is concentrated in the clocks and their disagreements: the anchor and its seven disagreement features (`e_*`) carry 14.6%, and the model leans on how far the schedule, the ADS-B first point and the best NM clock lie from the anchor (Table 12, Figure 11). Identity follows, mostly the airport, the stand and the flag that says whether the aircraft type was filled, which marks departures without an NM record. The single most useful traffic feature is the age of the last arrival at the same stand, a proxy for whether the aircraft has just turned around. Weather spreads 6.6% over 61 features, in line with Section 2.6: rare conditions matter for the flights they touch, not for the average.

Table 12: Share of model-A importance by feature group.

Group	Features	Importance
NM clocks and anchor	35	**32.4%**
Flight identity	14	**18.9%**
Airport traffic	71	**16.9%**
ADS-B evidence	58	**10.4%**
EUROCONTROL statistics	45	7.2%
Schedule and calendar	18	6.8%
Weather	61	6.6%
OPDI events	3	**0.7%**

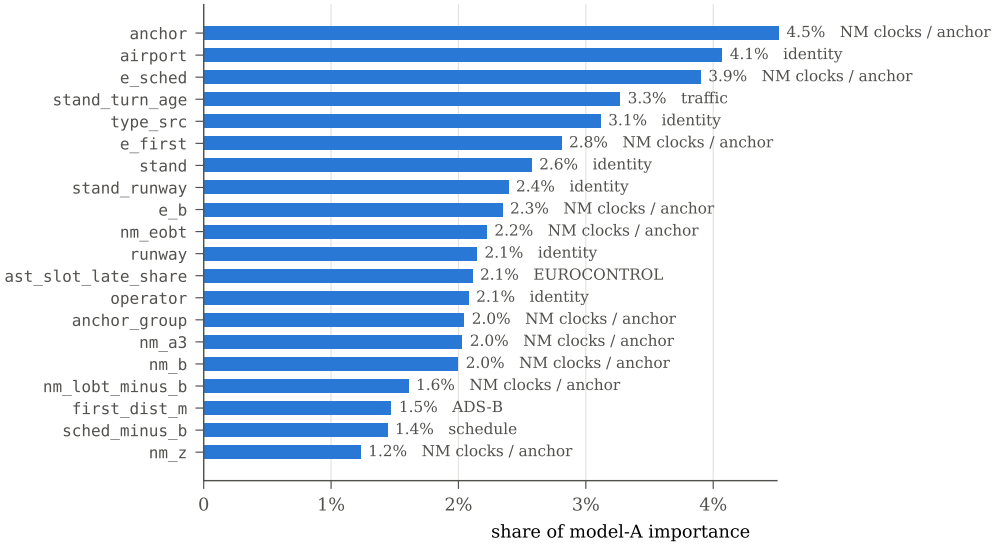


Figure 11: The 20 most important model-A features, with their group (share of CatBoost's prediction-value change, mean of the four seeds of the final pipeline).

7. Conclusion

Taxi-out time at these airports is a timestamp-reconciliation problem before it is a surface-dynamics problem. A ridge anchor on the flight-plan clocks, corrected by ADS-B where the aircraft was heard and by a residual stack elsewhere, reaches 124 s where the pushback is observed and 110–147 s at Schiphol, Zurich, Frankfurt and Munich. The score is set by the 1–2% of departures without a flight plan, above all at Rome, and by airports whose off-block clocks are rounded or unobserved. Waiting at airport hotspots costs roughly a minute of taxi-out per minute stopped but explains only part of the variation. The model versions reached 227.2 s on the ranking set with layers outside the pipeline described here, and 229.1 s with the pipeline's models only, after the final phase added airport-disruption features, eight seeds and an ensemble with the earlier stack, each chosen on the 2025 labels and confirmed on January and July 2026. For the final set, which adds February and June, those months are predicted by the model, and January and July carry the score-weighted ensemble (Section 6.5). At Rome the limit is not the model: the stamped-or-ordinary decision is a coin flip on a third of the flights by every signal in the data, so a calibrated hedge, about 3,000 s on that population, is

the ceiling without an independent record of the off-block. The 212.8 s of the score-weighted ensemble is a property of the ranking set, not of the method. Code: `Somnathab3/strong-crane` (GPLv3).

Use of AI tools

Claude (Anthropic) [5] was used for coding and for research. The author chose the methods, decided which results to keep, checked every number against the code outputs and takes full responsibility for the content.

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